EXPERIMENT 23

NAME: B.Akshay kumar Reddy

REG NO:192412246

CODE:

```
clc;
clear;
close all;
% Parameters
M = 16;                % 16-QAM
N = 1000;              % Number of symbols
data = randi([0 M-1],N,1);
% QAM Modulation
tx = qammod(data,M,'UnitAveragePower',true);
% AWGN Channel
rx = awgn(tx,10,'measured');    % 10 dB SNR
% Demodulation
data_rx = qamdemod(rx,M,'UnitAveragePower',true);
% BER Calculation
[num, ber] = biterr(data,data_rx);
% Display BER
fprintf('Bit Error Rate (BER) = %f\n',ber);
% Eye Diagram Signal
eyeSig = repelem(real(rx),8);
% Plots
figure;
subplot(2,2,1)
```

```
scatter(real(tx),imag(tx),'filled')
title('16-QAM Transmitted Constellation')
grid on
subplot(2,2,2)
scatter(real(rx),imag(rx),'filled')
title('Received Constellation (AWGN)')
grid on
subplot(2,2,[3 4])
eyediagram(eyeSig,16)
title('Eye Diagram')
```

OUTPUT:

